CS 2470 HW I Minglang Yin

1. Perceptron Learning algorithm updates the weights & bias for every training data, however our algorithm update using minibatch. And ② Perceptron learning algorithm typically works for binary classification, not for more complicated case.

2. To derive $\frac{\partial L}{\partial w_{ji}}$.

$\frac{\partial L}{\partial w_{ji}} = \frac{\partial L}{\partial P_c} \cdot \frac{\partial P_c}{\partial l_j} \cdot \frac{\partial l_j}{\partial w_{ji}}$, where $\frac{\partial L}{\partial P_c} = -\frac{1}{P_c}$, $\frac{\partial l_j}{\partial w_{ji}} = \frac{\partial(\sum_k w_{jk} x_k + b_j)}{\partial w_{ji}} = x_i$

$\frac{\partial P_c}{\partial l_j} = \frac{\partial\left(\frac{e^{l_c}}{\sum_k e^{l_k}}\right)}{\partial l_j} = \begin{cases} j = c: & \frac{e^{l_c}(\sum_k e^{l_k}) - e^{l_c} e^{l_c}}{(\sum_k e^{l_k})^2} = \frac{e^{l_j}}{\sum_k e^{l_k}} \cdot \frac{\sum_k e^{l_k} - e^{l_j}}{\sum_k e^{l_k}} = (1 - P_j) P_c \\ j \neq c: & \frac{-e^{l_c} e^{l_j}}{(\sum_k e^{l_k})^2} = -P_j P_c \end{cases}$

$\frac{\partial L}{\partial w_{ji}} = -\frac{1}{P_c}(y - P_j) P_c x_i = -(y - P_j) x_i$

3. Answer: ② Bias can be seen as an offset representing the apriori historical likeliness of being a certain class. The reason of adding bias is two-fold:
(1) ensure a non-zero output.
(2) it helps to fit the data

4: Answer: GPU is a kind of unit which is suitable for massive, simple parallel computing. Since training a deep neural network needs to perform repetively the tensor operation ② and grad·backward propagation, where they can be computed easily on GPU, hence GPU is invaluable in training neural networks.

2 Ethical Implications

1: MNIST is a pre-processed large classification dataset with recorded human ~~black & white~~ test & computer algorithm test. The images inside are ~~to~~ different but can be classified into digits. providing an available library for testing. Hence it is good for classification problem.

2: (a) The identification of zip code would be much faster. It would also reduce the human error for wrong classification

(b) Downside: ~~Post~~ Some Postman would not be happy since they would lost their job. Also, it would ~~incr~~ increase the cost spent on maintaining this system.